Guided ImMAP ascent:
one gradient step, a chain of conditional expectations

ImMAP project notes

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Abstract

We want a target image $\boldsymbol{x}_0$ given side information: a paired source image $\boldsymbol{x}_1$, a measurement $\boldsymbol{y}$, or both. The log-posterior splits along any order in which the side information is revealed. It becomes the target prior plus one pointwise mutual information (PMI) term per revealed variable, conditional on what was revealed before. This split survives Gaussian smoothing exactly. By Tweedie's formula, the gradient of each PMI term is the difference of the two conditional expectations it links. So one ascent step, $\boldsymbol{u}^+ = (1-h)\boldsymbol{u} + h\boldsymbol{v}$, handles any combination of guide and measurement. The estimate $\boldsymbol{v} = D_{\boldsymbol{c}_0} + \sum_k w_k (D_{\boldsymbol{c}_k} - D_{\boldsymbol{c}_{k-1}})$ is a weighted telescoping sum of denoisers $D_{\boldsymbol{c}}(\boldsymbol{u}) = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{c}]$. The algorithms below assume nothing about how each expectation is computed: a trained network, an approximation, or an oracle. We give the weights that guarantee ascent, and show that I^{2}SB, I^{2}SB-Recon, SDEdit and classifier-free guidance differ only in the conditioning chain, the weights and the step size.

Setting. Real vectors in $\mathbb{R}^n$; the complex case follows as in the main note on ImMAP-SB. Pairs $(\boldsymbol{x}_0, \boldsymbol{x}_1)$ have joint law $p(\boldsymbol{x}_0, \boldsymbol{x}_1)$. The measurement is $\boldsymbol{y} = \boldsymbol{E}\boldsymbol{x}_0 + \boldsymbol{n}$ with $\boldsymbol{n}$ independent of $(\boldsymbol{x}_0, \boldsymbol{x}_1)$; nothing below uses its form. φ_σ is the $\mathcal{N}(\boldsymbol{0}, \sigma^2 \boldsymbol{I})$ density and $\boldsymbol{u} = \boldsymbol{x}_0 + \sigma\boldsymbol{\varepsilon}$, with $\boldsymbol{\varepsilon}$ independent of $(\boldsymbol{x}_0, \boldsymbol{x}_1, \boldsymbol{y})$. On the bridge, $\boldsymbol{u}$ is the debiased state $(\boldsymbol{x}_t - t\boldsymbol{x}_1)/(1-t)$ at $\sigma^2 = Vt/(1-t)$.

Conditional expectations. For any conditioning variable $\boldsymbol{c}$ (a subset of $\{\boldsymbol{x}_1, \boldsymbol{y}\}$, possibly empty),

$$D_{\boldsymbol{c}}(\boldsymbol{u}) \coloneqq \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{c}], \qquad \text{at the current noise level } \sigma.$$

The four that appear are $D_\varnothing = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}]$, $D_{\boldsymbol{x}_1} = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1]$, $D_{\boldsymbol{y}} = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{y}]$ and $D_{\boldsymbol{x}_1,\boldsymbol{y}} = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1, \boldsymbol{y}]$. The derivation treats each as exact. In practice each is replaced by an estimator (Section 8), and the algorithms do not depend on which.

1 The objective along a conditioning chain

Reveal the side information in some order, giving a nested chain $\boldsymbol{c}_0 \subset \boldsymbol{c}_1 \subset \cdots \subset \boldsymbol{c}_K$ with $\boldsymbol{c}_0$ the variables that are always conditioned on (often none). Repeated Bayes gives the exact telescoping identity

$$\log p(\boldsymbol{x}_0 \mid \boldsymbol{c}_K) = \log p(\boldsymbol{x}_0 \mid \boldsymbol{c}_0) + \sum_{k=1}^{K} \underbrace{\log \frac{p(\boldsymbol{x}_0 \mid \boldsymbol{c}_k)}{p(\boldsymbol{x}_0 \mid \boldsymbol{c}_{k-1})}}_{\mathrm{PMI}(\boldsymbol{x}_0;\, \boldsymbol{c}_k \setminus \boldsymbol{c}_{k-1} \,\mid\, \boldsymbol{c}_{k-1})} . \tag{1}$$

Each term is the pointwise mutual information between the target and the newly revealed variable, given what was already known. We weight each term and maximise

$$\max_{\boldsymbol{x}_0} \ \log p(\boldsymbol{x}_0 \mid \boldsymbol{c}_0) + \sum_{k=1}^{K} w_k \ \mathrm{PMI}(\boldsymbol{x}_0; \boldsymbol{c}_k \setminus \boldsymbol{c}_{k-1} \mid \boldsymbol{c}_{k-1}), \tag{2}$$

where $w_k = 1$ for all k is the posterior given $\boldsymbol{c}_K$, $w_k = 0$ ignores the k-th variable, and $w_k > 1$ over-weights it as in classifier-free guidance [1]. The two chains used here:

- **No measurement:** $\varnothing \subset \{\boldsymbol{x}_1\}$, i.e. $\log p(\boldsymbol{x}_0) + w \ \mathrm{PMI}(\boldsymbol{x}_0; \boldsymbol{x}_1)$.
- **With measurement:** $\varnothing \subset \{\boldsymbol{x}_1\} \subset \{\boldsymbol{x}_1, \boldsymbol{y}\}$, i.e. $\log p(\boldsymbol{x}_0) + w \ \mathrm{PMI}(\boldsymbol{x}_0; \boldsymbol{x}_1) + w_y \ \mathrm{PMI}(\boldsymbol{x}_0; \boldsymbol{y} \mid \boldsymbol{x}_1)$. The other order, $\varnothing \subset \{\boldsymbol{y}\} \subset \{\boldsymbol{x}_1, \boldsymbol{y}\}$, gives the same posterior at unit weights but a different family when the weights differ.

2 Smoothing, and the gradient

As in ImMAP, we ascend the Gaussian smoothing of the objective while lowering the smoothing. Write $p_\sigma(\boldsymbol{u} \mid \boldsymbol{c}) = \int \varphi_\sigma(\boldsymbol{u} - \boldsymbol{x})\, p(\boldsymbol{x} \mid \boldsymbol{c})\, d\boldsymbol{x}$ for the law of the noisy target given $\boldsymbol{c}$, and

$$J_\sigma(\boldsymbol{u}) \coloneqq \log p_\sigma(\boldsymbol{u} \mid \boldsymbol{c}_0) + \sum_{k=1}^{K} w_k \log \frac{p_\sigma(\boldsymbol{u} \mid \boldsymbol{c}_k)}{p_\sigma(\boldsymbol{u} \mid \boldsymbol{c}_{k-1})}. \tag{3}$$

Proposition 1 (The chain survives smoothing)**.** *Each ratio in (3) is the PMI between the* noisy *target $\boldsymbol{u}$ and the newly revealed variable given $\boldsymbol{c}_{k-1}$. At unit weights $J_\sigma = \log p_\sigma(\boldsymbol{u} \mid \boldsymbol{c}_K)$, and $J_\sigma \to$ (2) as $\sigma \to 0$.*

Proof. Since $\boldsymbol{\varepsilon}$ is independent of all side information, Bayes applies to $(\boldsymbol{u}, \boldsymbol{c}_k)$ exactly as to $(\boldsymbol{x}_0, \boldsymbol{c}_k)$, and each ratio equals $p(\boldsymbol{c}_k \setminus \boldsymbol{c}_{k-1} \mid \boldsymbol{u}, \boldsymbol{c}_{k-1}) / p(\boldsymbol{c}_k \setminus \boldsymbol{c}_{k-1} \mid \boldsymbol{c}_{k-1})$. At unit weights the sum telescopes. As $\sigma \to 0$, $p_\sigma(\cdot \mid \boldsymbol{c}) \to p(\cdot \mid \boldsymbol{c})$. □

The smoothed terms are not the unsmoothed ones evaluated at a noisy image. For the measurement, for instance, $p(\boldsymbol{y} \mid \boldsymbol{u}, \boldsymbol{x}_1) = \int p(\boldsymbol{y} \mid \boldsymbol{x})\, p(\boldsymbol{x} \mid \boldsymbol{u}, \boldsymbol{x}_1)\, d\boldsymbol{x}$: the likelihood averaged over the targets that the noisy state still allows. It is weak at large σ and sharpens to $p(\boldsymbol{y} \mid \boldsymbol{u})$ as $\sigma \to 0$.

Proposition 2 (Each information term is a difference of expectations)**.**

$$\sigma^2 \nabla J_\sigma(\boldsymbol{u}) = \boldsymbol{v} - \boldsymbol{u}, \qquad \boldsymbol{v} = D_{\boldsymbol{c}_0}(\boldsymbol{u}) + \sum_{k=1}^{K} w_k \left(D_{\boldsymbol{c}_k}(\boldsymbol{u}) - D_{\boldsymbol{c}_{k-1}}(\boldsymbol{u}) \right). \tag{4}$$

Proof. Tweedie's formula, $\sigma^2 \nabla \log p_\sigma(\boldsymbol{u} \mid \boldsymbol{c}) = D_{\boldsymbol{c}}(\boldsymbol{u}) - \boldsymbol{u}$, holds for any $\boldsymbol{c}$ independent of $\boldsymbol{\varepsilon}$ (differentiate the convolution under the integral). Apply it to every log-density in (3); the $-\boldsymbol{u}$ terms cancel inside each ratio. □

The information a variable carries about the target, at noise level σ, is how far revealing it moves the conditional expectation. The term vanishes when the variable is uninformative given the rest. Equivalently, regrouping (4) with $w_0 \coloneqq 1$ and $w_{K+1} \coloneqq 0$,

$$\boldsymbol{v} = \sum_{k=0}^{K} a_k \, D_{\boldsymbol{c}_k}(\boldsymbol{u}), \qquad a_k = w_k - w_{k+1}, \qquad \sum_k a_k = 1, \tag{5}$$

an affine combination of the conditional expectations along the chain. The two chains give

$$\text{no measurement:} \quad \boldsymbol{v} = D_{\varnothing} + w\,(D_{\boldsymbol{x}_1} - D_{\varnothing}) = (1-w)\,D_{\varnothing} + w\,D_{\boldsymbol{x}_1}, \tag{6}$$

$$\begin{aligned}\text{with measurement:} \quad \boldsymbol{v} &= D_{\varnothing} + w\,(D_{\boldsymbol{x}_1} - D_{\varnothing}) + w_y\,(D_{\boldsymbol{x}_1,\boldsymbol{y}} - D_{\boldsymbol{x}_1}) \\ &= (1-w)\,D_{\varnothing} + (w-w_y)\,D_{\boldsymbol{x}_1} + w_y\,D_{\boldsymbol{x}_1,\boldsymbol{y}}.\end{aligned} \tag{7}$$

At unit weights only the last expectation survives: $\boldsymbol{v} = D_{\boldsymbol{x}_1}$ without a measurement and $\boldsymbol{v} = D_{\boldsymbol{x}_1,\boldsymbol{y}}$ with one.

3 The gradient step, and when it ascends

Gradient ascent on J_σ with step size $h\sigma^2$ is, by (4),

$$\boxed{\boldsymbol{u}^+ = \boldsymbol{u} + h\sigma^2 \nabla J_\sigma(\boldsymbol{u}) = (1-h)\,\boldsymbol{u} + h\,\boldsymbol{v}.} \tag{8}$$

Written out, the step moves along a sum of independent pulls, one per term of the objective:

$$\boldsymbol{u}^+ - \boldsymbol{u} = h\,(D_{\boldsymbol{c}_0} - \boldsymbol{u}) + \sum_{k=1}^{K} h\,w_k\,(D_{\boldsymbol{c}_k} - D_{\boldsymbol{c}_{k-1}}),$$

where the first pulls toward the always-known conditional mean, and the k-th by what the k-th variable adds.

Proposition 3 (Ascent)**.** *Let R bound the support radius of $p(\boldsymbol{x}_0 \mid \boldsymbol{c}_0)$, and let a_k be as in (5). Then $-\nabla^2 J_\sigma \preceq L\boldsymbol{I}$ with*

$$L = \frac{\sigma^2 + R^2 \sum_{k:\,a_k<0} |a_k|}{\sigma^4}, \tag{9}$$

and every step with $0 < h\sigma^2 < 2/L$ increases J_σ: $J_\sigma(\boldsymbol{u}^+) \geq J_\sigma(\boldsymbol{u}) + h\sigma^2(1 - \frac{1}{2}h\sigma^2 L)\|\nabla J_\sigma(\boldsymbol{u})\|^2$. In particular, if the weights are non-increasing along the chain, $1 \geq w_1 \geq \cdots \geq w_K \geq 0$, then all $a_k \geq 0$, $L = \sigma^{-2}$, and every $h \in (0,2)$ ascends.

Proof. By (3), $J_\sigma = \sum_k a_k \log p_\sigma(\boldsymbol{u} \mid \boldsymbol{c}_k)$. Tweedie's second-order identity gives $\nabla^2 \log p_\sigma(\boldsymbol{u} \mid \boldsymbol{c}) = (C_{\boldsymbol{c}} - \sigma^2\boldsymbol{I})/\sigma^4$ with $C_{\boldsymbol{c}} = \mathrm{Cov}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{c}] \succeq 0$. Hence $-\nabla^2 J_\sigma = (\sigma^2\boldsymbol{I} - \sum_k a_k C_{\boldsymbol{c}_k})/\sigma^4$, using $\sum_k a_k = 1$. Drop the terms with $a_k \geq 0$ and bound $C_{\boldsymbol{c}_k} \preceq R^2\boldsymbol{I}$ in the others. The inequality is the descent lemma along $h\sigma^2\nabla J_\sigma$. □

Over-weighting a variable ($w_k > w_{k-1}$) makes some a_k negative: J_σ then pushes *away* from a less-conditioned expectation. The safe step shrinks by $R^2 \sum |a_k|/\sigma^2$, most at small σ, so a guidance weight is best kept for the large-σ part of the anneal.

4 The algorithms

The step (8) runs inside the ImMAP continuation [2]. After each step, fresh noise is injected so that the iterate is again a noisy target at the next, smaller level, and σ is lowered. With the contraction $\sigma' = (1-\beta h)\sigma$, $\beta \in (0,1]$, an accurate step leaves noise $(1-h)\sigma\boldsymbol{\varepsilon}$, so the injection is

$$\gamma = \sqrt{(1-\beta h)^2 - (1-h)^2}\;\sigma. \tag{10}$$

$\beta = 1$ injects nothing and ascends deterministically toward the MAP estimate; $\beta < 1$ samples. Algorithm 1 is the general form. Algorithms 2 and 3 are its two instances, written out. Every D is a call to whatever estimates that conditional expectation.

Algorithm 1 Guided ImMAP ascent, general conditioning chain

Require: chain $\boldsymbol{c}_0 \subset \cdots \subset \boldsymbol{c}_K$; estimators of $D_{\boldsymbol{c}_k}(\boldsymbol{u}) = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{c}_k]$; weights $w_1, \ldots, w_K$; start $\boldsymbol{u}_0$ at level σ_0; $\beta \in (0, 1]$; steps h_k (Prop. 3); $\sigma_{\min}$

$\boldsymbol{u} \leftarrow \boldsymbol{u}_0$, $\sigma \leftarrow \sigma_0$
while $\sigma > \sigma_{\min}$ **do**
  $\boldsymbol{v} \leftarrow D_{\boldsymbol{c}_0}(\boldsymbol{u})$ ▷ $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{c}_0]$
  for $k = 1, \ldots, K$ **do**
    $\boldsymbol{v} \leftarrow \boldsymbol{v} + w_k \left(D_{\boldsymbol{c}_k}(\boldsymbol{u}) - D_{\boldsymbol{c}_{k-1}}(\boldsymbol{u})\right)$ ▷ $+ w_k \sigma^2 \nabla \mathrm{PMI}_\sigma(\boldsymbol{u}; \boldsymbol{c}_k \setminus \boldsymbol{c}_{k-1} \mid \boldsymbol{c}_{k-1})$
  end for ▷ now $\boldsymbol{v} = \boldsymbol{u} + \sigma^2 \nabla J_\sigma(\boldsymbol{u})$
  $\boldsymbol{u} \leftarrow (1 - h)\, \boldsymbol{u} + h\, \boldsymbol{v} + \gamma\, \boldsymbol{z}$, $\boldsymbol{z} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$ ▷ step (8), injection (10)
  $\sigma \leftarrow (1 - \beta h)\, \sigma$
end while
return $\boldsymbol{v}$

Expectations with weight-difference $a_k = 0$ need not be evaluated: at unit weights the loop reduces to the single call $\boldsymbol{v} \leftarrow D_{\boldsymbol{c}_K}(\boldsymbol{u})$.

Algorithm 2 No measurement: maximise $\log p(\boldsymbol{x}_0) + w\ \mathrm{PMI}(\boldsymbol{x}_0; \boldsymbol{x}_1)$

Require: source $\boldsymbol{x}_1$; estimators of $D_\varnothing$ and $D_{\boldsymbol{x}_1}$; weight w; $\boldsymbol{u}_0$, σ_0, β, h_k, $\sigma_{\min}$

$\boldsymbol{u} \leftarrow \boldsymbol{u}_0$, $\sigma \leftarrow \sigma_0$
while $\sigma > \sigma_{\min}$ **do**
  $\boldsymbol{v}_\varnothing \leftarrow D_\varnothing(\boldsymbol{u})$ ▷ $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}]$ (skip if $w = 1$)
  $\boldsymbol{v}_1 \leftarrow D_{\boldsymbol{x}_1}(\boldsymbol{u})$ ▷ $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1]$
  $\boldsymbol{v} \leftarrow \boldsymbol{v}_\varnothing + w\, (\boldsymbol{v}_1 - \boldsymbol{v}_\varnothing)$ ▷ prior pull + $w\times$ guide information
  $\boldsymbol{u} \leftarrow (1 - h)\, \boldsymbol{u} + h\, \boldsymbol{v} + \gamma\, \boldsymbol{z}$; $\sigma \leftarrow (1 - \beta h)\, \sigma$
end while
return $\boldsymbol{v}$

Algorithm 3 With measurement: maximise $\log p(\boldsymbol{x}_0) + w\ \mathrm{PMI}(\boldsymbol{x}_0; \boldsymbol{x}_1) + w_y\ \mathrm{PMI}(\boldsymbol{x}_0; \boldsymbol{y} \mid \boldsymbol{x}_1)$

Require: source $\boldsymbol{x}_1$; measurement $\boldsymbol{y}$; estimators of $D_\varnothing$, $D_{\boldsymbol{x}_1}$ and $D_{\boldsymbol{x}_1, \boldsymbol{y}}$; weights w, w_y; $\boldsymbol{u}_0$, σ_0, β, h_k, $\sigma_{\min}$

$\boldsymbol{u} \leftarrow \boldsymbol{u}_0$, $\sigma \leftarrow \sigma_0$
while $\sigma > \sigma_{\min}$ **do**
  $\boldsymbol{v}_\varnothing \leftarrow D_\varnothing(\boldsymbol{u})$ ▷ $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}]$ (skip if $w = 1$)
  $\boldsymbol{v}_1 \leftarrow D_{\boldsymbol{x}_1}(\boldsymbol{u})$ ▷ $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1]$ (skip if $w = w_y$)
  $\boldsymbol{v}_{1y} \leftarrow D_{\boldsymbol{x}_1, \boldsymbol{y}}(\boldsymbol{u})$ ▷ $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1, \boldsymbol{y}]$
  $\boldsymbol{v} \leftarrow \boldsymbol{v}_\varnothing + w\, (\boldsymbol{v}_1 - \boldsymbol{v}_\varnothing) + w_y\, (\boldsymbol{v}_{1y} - \boldsymbol{v}_1)$ ▷ prior + guide + measurement information
  $\boldsymbol{u} \leftarrow (1 - h)\, \boldsymbol{u} + h\, \boldsymbol{v} + \gamma\, \boldsymbol{z}$; $\sigma \leftarrow (1 - \beta h)\, \sigma$
end while
return $\boldsymbol{v}$

The two algorithms share the step, the injection and the anneal. They differ only in how many

conditional expectations enter $\boldsymbol{v}$, and with which weights. A natural start is $\boldsymbol{u}_0 = D_{\boldsymbol{c}_K}$ evaluated at $\sigma = \infty$ (e.g. $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_1]$), re-noised to σ_0.

5 I^{2}SB as an instance

I^{2}SB [3] is Algorithm 2 with every choice fixed. Two are about *what* is estimated, the rest are hyperparameters.

Conditioning. The chain is $\varnothing \subset \{\boldsymbol{x}_1\}$ with $w = 1$, so by (6) $\boldsymbol{v} = D_{\boldsymbol{x}_1}(\boldsymbol{u}) = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1]$. The target-only expectation $D_\varnothing$ has weight $1 - w = 0$ and is never evaluated. The objective is the conditional prior $\log p(\boldsymbol{x}_0 \mid \boldsymbol{x}_1)$; nothing is over- or under-weighted.

Estimator. $D_{\boldsymbol{x}_1}$ is the paired bridge regressor, trained with the squared loss. It is called on the bridge state: $D_{\boldsymbol{x}_1}(\boldsymbol{u}) = D_\theta(\boldsymbol{x}_t, t, \boldsymbol{x}_1)$ with $\boldsymbol{x}_t = (1-t)\boldsymbol{u} + t\boldsymbol{x}_1$ and $t = \sigma^2/(\sigma^2 + V)$.

Noise ladder. Fixed in advance: NFE $+ 1$ bridge times $1 = t_0 > t_1 > \cdots > t_K = 0$, uniform in the discrete step index, and $\sigma_k = \sigma(t_k)$, $\sigma^2(t) = Vt/(1-t)$. So the ladder runs from $\sigma_0 = \infty$ to $\sigma_K = 0$, and $V = 4\tau^2$ sets its scale.

Step size and injection. With $r_k = \sigma_{k+1}/\sigma_k$, the step removes exactly the variance between consecutive rungs, and β is set so that the contraction lands on the next rung:

$$h_k = 1 - r_k^2 \quad (h_k \sigma_k^2 = \sigma_k^2 - \sigma_{k+1}^2), \qquad \beta_k = \frac{1}{1 + r_k}, \qquad \gamma_k^2 = \sigma_{k+1}^2 \,(1 - r_k^2). \tag{11}$$

Proposition 4 (I^{2}SB's reverse step)**.** *With (11), one step of Algorithm 2 at $w = 1$ is*

$$\boldsymbol{u}_{k+1} = r_k^2 \, \boldsymbol{u}_k + (1 - r_k^2)\, D_{\boldsymbol{x}_1}(\boldsymbol{u}_k) + \sigma_{k+1}\sqrt{1 - r_k^2}\, \boldsymbol{z},$$

which in bridge coordinates is I^{2}SB's reverse (DDPM-posterior) step. At the two ends: the first step ($\sigma_0 = \infty$, $r_0 = 0$) is $\boldsymbol{u}_1 = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_1] + \sigma_1 \boldsymbol{z}$, so I^{2}SB's "start at $\boldsymbol{x}_1$" is the warm start of Algorithm 2, and the starting $\boldsymbol{u}_0$ is irrelevant. The last step ($\sigma_K = 0$, $r_{K-1} = 0$) returns $\boldsymbol{v} = D_{\boldsymbol{x}_1}(\boldsymbol{u}_{K-1})$ with no injected noise.

Proof. $1 - \beta_k h_k = 1 - (1 - r_k)(1 + r_k)/(1 + r_k) = r_k$, so $\sigma_{k+1} = r_k \sigma_k$ as required, and $\gamma_k^2 = [r_k^2 - r_k^4]\sigma_k^2 = \sigma_{k+1}^2(1 - r_k^2)$. Substituting in (8) gives the displayed step. Mapping both sides to $\boldsymbol{x} = (1-t)\boldsymbol{u} + t\boldsymbol{x}_1$ gives I^{2}SB's posterior mean $(1 - \frac{s}{t})D_{\boldsymbol{x}_1} + \frac{s}{t}\boldsymbol{x}_t$ and variance $V\frac{s(t-s)}{t}$, as shown in the main note (proposition *I^{2}SB*). At $r = 0$: $h = 1$ removes the current state entirely, and $\gamma = \sigma_{k+1}$. □

I^{2}SB-Recon. The same, one level further down the chain: $\varnothing \subset \{\boldsymbol{x}_1\} \subset \{\boldsymbol{x}_1, \boldsymbol{y}\}$ with $w = w_y = 1$, so $\boldsymbol{v} = D_{\boldsymbol{x}_1, \boldsymbol{y}} = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1, \boldsymbol{y}]$, with the ladder, step sizes and injection of (11). The only new choice is the estimator: $D_{\boldsymbol{x}_1, \boldsymbol{y}}$ is obtained from $D_{\boldsymbol{x}_1}$ by 7 CG iterations on the proximal problem with $\lambda = 7 \cdot 10^{-5}$ (Section 8). When $\boldsymbol{x}_1 = \boldsymbol{E}^H \boldsymbol{y}$ (the zero-filled start), $\boldsymbol{x}_1$ is a sufficient statistic for $\boldsymbol{x}_0$ given $\boldsymbol{y}$, so $D_{\boldsymbol{x}_1, \boldsymbol{y}} = D_{\boldsymbol{x}_1}$ exactly. Then the measurement term $\mathrm{PMI}(\boldsymbol{x}_0; \boldsymbol{y} \mid \boldsymbol{x}_1)$ vanishes, and the CG step corrects only the network's errors.

6 ImMAP2 as an instance

ImMAP2 (`diffusion/immap.py`, `immap2`) is Algorithm 1 on a different chain, with the ImMAP choices of hyperparameters.

Conditioning. There is no guide. The chain is $\varnothing \subset \{\boldsymbol{y}\}$ with $w_y = 1$, so $\boldsymbol{v} = D_{\boldsymbol{y}}(\boldsymbol{u}) = \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{y}]$, and the objective is the posterior $\log p(\boldsymbol{x}_0 \mid \boldsymbol{y})$.

Estimator. $D_{\boldsymbol{y}}$ is obtained from the target-only denoiser $D_\varnothing$ by the isotropic Gaussian projection of Section 8, applied to $D_\varnothing$ instead of $D_{\boldsymbol{x}_1}$:

$$D_{\boldsymbol{y}}(\boldsymbol{u}) \approx (\boldsymbol{E}^H\boldsymbol{E} + \lambda_k \boldsymbol{I})^{-1}(\boldsymbol{E}^H\boldsymbol{y} + \lambda_k D_\varnothing(\boldsymbol{u})), \qquad \lambda_k = \lambda_0\, \sigma_n^2\, \frac{1+\sigma_k^2}{\sigma_k^2}. \tag{12}$$

This is the prox with posterior variance $\varsigma_k^2 = \sigma_k^2/(1+\sigma_k^2)$, scaled by $1/\lambda_0$. That is the posterior variance of a unit-variance Gaussian signal observed at noise σ_k (the ΠGDM rule), so λ_k follows the noise level. It is solved by CG essentially to convergence (up to 500 iterations, tolerance 10^{-3}).

Noise ladder. Self-paced rather than fixed. Each iteration re-estimates the level from the denoiser residual,

$$\sigma_k^2 = \tfrac{1}{n}\, \|D_\varnothing(\boldsymbol{u}_k) - \boldsymbol{u}_k\|^2, \tag{13}$$

which is close to σ_k^2 when the denoiser's error is small next to the noise: by the orthogonality of the MMSE error, $\mathbb{E}\|D_\varnothing(\boldsymbol{u}) - \boldsymbol{u}\|^2 = n\sigma^2 - \mathrm{mmse}(\sigma)$. It replaces the open-loop contraction $\sigma \leftarrow (1-\beta h)\sigma$ of Algorithm 1. The iteration stops once $\sigma_k < \sigma_{\min} = 0.01$.

Step size and injection. The Kadkhodaie–Simoncelli schedule [2] and a fixed, small β:

$$h_k = \frac{h_0\, k}{1 + h_0(k-1)}, \quad h_0 = 0.01, \qquad \beta = 0.05, \qquad \gamma_k = \sigma_k \sqrt{(1-\beta h_k)^2 - (1-h_k)^2}. \tag{14}$$

The step starts tiny and grows toward 1. $\beta = 0.05$ re-injects most of the noise each step removes, so the iteration is a slow, strongly stochastic ascent.

Start and output. $\boldsymbol{u}_0 = \mathrm{mean}(\boldsymbol{E}^H\boldsymbol{y})\,\boldsymbol{1} + \sigma_{\max}\boldsymbol{z}$ with $\sigma_{\max} = 1$, or a warm start $\boldsymbol{u}_0 = \boldsymbol{x}_{\mathrm{ws}} + \sigma_{\max}\boldsymbol{z}$. The implementation returns the final state $\boldsymbol{u}$.

Algorithm 4 ImMAP2 as an instance: chain $\varnothing \subset \{\boldsymbol{y}\}$, $w_y = 1$

Require: $\boldsymbol{y}, \boldsymbol{E}, \sigma_n$; target-only denoiser $D_\varnothing$; λ_0; $h_0 = 0.01$; $\beta = 0.05$; $\sigma_{\max} = 1$; $\sigma_{\min} = 0.01$

$\boldsymbol{u} \leftarrow \mathrm{mean}(\boldsymbol{E}^H\boldsymbol{y})\,\boldsymbol{1} + \sigma_{\max}\boldsymbol{z}$; $\sigma \leftarrow \sigma_{\max}$; $k \leftarrow 1$

while $\sigma > \sigma_{\min}$ **do**

  $\boldsymbol{v}_\varnothing \leftarrow D_\varnothing(\boldsymbol{u})$ ▷ $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}]$

  $\sigma^2 \leftarrow \frac{1}{n}\|\boldsymbol{v}_\varnothing - \boldsymbol{u}\|^2$ ▷ self-paced level (13)

  $\boldsymbol{v} \leftarrow (\boldsymbol{E}^H\boldsymbol{E} + \lambda_k\boldsymbol{I})^{-1}(\boldsymbol{E}^H\boldsymbol{y} + \lambda_k\boldsymbol{v}_\varnothing)$ ▷ $\approx \mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{y}]$, (12)

  $h_k \leftarrow h_0 k/(1 + h_0(k-1))$; $\gamma \leftarrow \sigma\sqrt{(1-\beta h_k)^2 - (1-h_k)^2}$

  $\boldsymbol{u} \leftarrow (1-h_k)\,\boldsymbol{u} + h_k\,\boldsymbol{v} + \gamma\,\boldsymbol{z}$; $k \leftarrow k+1$ ▷ step (8)

end while

return $\boldsymbol{u}$

I^{2}SB-Recon and ImMAP2 side by side. Both maximise a smoothed posterior with the same step (8). Every difference is a choice of conditioning or a hyperparameter:

choice	I^{2}SB-Recon	ImMAP2
chain, weights	$\varnothing \subset \{\boldsymbol{x}_1\} \subset \{\boldsymbol{x}_1, \boldsymbol{y}\}$, $w = w_y = 1$	$\varnothing \subset \{\boldsymbol{y}\}$, $w_y = 1$
$\boldsymbol{v}$ estimates	$\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1, \boldsymbol{y}]$	$\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{y}]$
denoiser	paired bridge regressor $D_{\boldsymbol{x}_1}$	target-only $D_\varnothing$
data estimator	prox from $D_{\boldsymbol{x}_1}$, $\lambda = 7 \cdot 10^{-5}$ fixed, 7 CG iterations	prox from $D_\varnothing$, $\lambda_k = \lambda_0 \sigma_n^2 (1 + \sigma_k^2)/\sigma_k^2$, CG to tolerance
noise ladder	fixed: $\sigma(t_k)$ on NFE uniform steps, $\infty \to 0$	self-paced: residual (13), $1 \to 0.01$
step size h_k	$1 - \sigma_{k+1}^2/\sigma_k^2$ (variance removed)	$h_0 k/(1 + h_0(k-1))$, $h_0 = 0.01$
injection β	$1/(1 + r_k) \approx \frac{1}{2}$	0.05
start	$\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{x}_1] + \sigma_1 \boldsymbol{z}$ (first step)	mean$(\boldsymbol{E}^H \boldsymbol{y}) + \boldsymbol{z}$, or warm start
iterations	NFE (e.g. 20)	until $\sigma < 0.01$ (hundreds)

Table 1: The two methods as instances of Algorithm 1. The first three rows say what is estimated; the rest are hyperparameters.

Moving along a row turns one method into the other, which gives a set of controlled comparisons. Give ImMAP2 the guide (chain $\varnothing \subset \{\boldsymbol{x}_1\} \subset \{\boldsymbol{x}_1, \boldsymbol{y}\}$ with $D_{\boldsymbol{x}_1}$ in place of $D_\varnothing$) and it becomes ImMAP-SB with data. Give I^{2}SB-Recon ImMAP2's λ_k and the gap between them is the effect of the data weight alone. To interpolate between the two targets, reveal the measurement first: on the chain $\varnothing \subset \{\boldsymbol{y}\} \subset \{\boldsymbol{x}_1, \boldsymbol{y}\}$ with $w_y = 1$, $\boldsymbol{v} = D_{\boldsymbol{y}} + w\,(D_{\boldsymbol{x}_1, \boldsymbol{y}} - D_{\boldsymbol{y}})$, which is ImMAP2's $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{y}]$ at $w = 0$ and I^{2}SB-Recon's $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1, \boldsymbol{y}]$ at $w = 1$. (On the other order, $\varnothing \subset \{\boldsymbol{x}_1\} \subset \{\boldsymbol{x}_1, \boldsymbol{y}\}$, $w = 0$ does not give ImMAP2: the chain order matters once the weights differ.)

7 What each method is

method	chain	weights	$\boldsymbol{v}$ estimates	step, noise
SDEdit, ImMAP sampler	$\varnothing \subset \{\boldsymbol{x}_1\}$	$w = 0$	$\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}]$	free
ImMAP2 (Sec. 6)	$\varnothing \subset \{\boldsymbol{y}\}$	$w_y = 1$	$\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{y}]$	KS h_k, $\beta = 0.05$
I^{2}SB (Sec. 5)	$\varnothing \subset \{\boldsymbol{x}_1\}$	$w = 1$	$\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1]$	$h = 1 - r^2$, $\beta = \frac{1}{1+r}$
ImMAP-SB	$\varnothing \subset \{\boldsymbol{x}_1\}$	$w = 1$	$\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1]$	free
guided (CFG on the guide)	$\varnothing \subset \{\boldsymbol{x}_1\}$	$w > 1$	$(1 - w)D_\varnothing + wD_{\boldsymbol{x}_1}$	Prop. 3
I^{2}SB-Recon, ImMAP-SB with data, SB-TGVN	$\varnothing \subset \{\boldsymbol{x}_1\} \subset \{\boldsymbol{x}_1, \boldsymbol{y}\}$	$w = w_y = 1$	$\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1, \boldsymbol{y}]$	I^{2}SB's or free
guided recon	same	$w \neq 1$ and/or $w_y \neq 1$	(7)	Prop. 3

Table 2: Every method is the step (8) with a different chain and weights. The methods sharing a row differ only in how they *estimate* the same expectation (Section 8). "Free" is the ImMAP choice of h_k and β.

Ingredients for an experiment. Varying w needs $D_\varnothing$ and $D_{\boldsymbol{x}_1}$ at the *same* noise level σ, in the *same* coordinates $\boldsymbol{u}$. A target-only denoiser (e.g. LPDS trained on PDFS over a range of σ) takes $\boldsymbol{u}$

directly. The paired bridge regressor is called on $\boldsymbol{x}_t = (1-t)\boldsymbol{u} + t\boldsymbol{x}_1$ at $t = \sigma^2/(\sigma^2 + V)$. The start σ_0 must lie inside the target denoiser's training range.

8 Estimating the expectations (outside the algorithm)

The algorithms consume estimates of $D_{\boldsymbol{c}}$ and are indifferent to their source. The current options, for reference:

- $D_{\varnothing}$: a target-only denoiser trained with the squared loss on $(\boldsymbol{u}, \boldsymbol{x}_0)$.
- $D_{\boldsymbol{x}_1}$: the paired bridge regressor, trained with the squared loss on bridge states.
- $D_{\boldsymbol{x}_1,\boldsymbol{y}}$, learned: a bridge regressor that also sees $(\boldsymbol{y}, \boldsymbol{E})$ (SB-TGVN, BridgeLGGS), trained toward exactly $\mathbb{E}[\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1, \boldsymbol{y}]$.
- $D_{\boldsymbol{x}_1,\boldsymbol{y}}$, derived from $D_{\boldsymbol{x}_1}$: if $p(\boldsymbol{x}_0 \mid \boldsymbol{u}, \boldsymbol{x}_1) \approx \mathcal{N}(D_{\boldsymbol{x}_1}, \boldsymbol{C})$ and $\boldsymbol{y} = \boldsymbol{E}\boldsymbol{x}_0 + \boldsymbol{n}$, $\boldsymbol{n} \sim \mathcal{N}(\boldsymbol{0}, \sigma_n^2 \boldsymbol{I})$, then Gaussian conditioning gives $D_{\boldsymbol{x}_1,\boldsymbol{y}} \approx D_{\boldsymbol{x}_1} + \boldsymbol{C}\boldsymbol{E}^H(\boldsymbol{E}\boldsymbol{C}\boldsymbol{E}^H + \sigma_n^2 \boldsymbol{I})^{-1}(\boldsymbol{y} - \boldsymbol{E}D_{\boldsymbol{x}_1})$. Tweedie's second-order identity supplies $\boldsymbol{C} = \sigma^2 \partial D_{\boldsymbol{x}_1}/\partial \boldsymbol{u}$. The isotropic choice $\boldsymbol{C} = c\sigma^2 \boldsymbol{I}$ reduces it to the proximal step $(\boldsymbol{E}^H\boldsymbol{E} + \lambda \boldsymbol{I})^{-1}(\boldsymbol{E}^H\boldsymbol{y} + \lambda D_{\boldsymbol{x}_1})$, $\lambda = \sigma_n^2/(c\sigma^2)$. I^{2}SB-Recon [4] uses that step with λ fixed at $7 \cdot 10^{-5}$ and 7 CG iterations.

References

[1] J. Ho, T. Salimans. Classifier-free diffusion guidance. *NeurIPS Workshop on Deep Generative Models and Downstream Applications*, 2021.

[2] Z. Kadkhodaie, E. P. Simoncelli. Stochastic solutions for linear inverse problems using the prior implicit in a denoiser. *NeurIPS*, 2021.

[3] G.-H. Liu, A. Vahdat, D.-A. Huang, E. A. Theodorou, W. Nie, A. Anandkumar. I^{2}SB: Image-to-Image Schrödinger Bridge. *ICML*, 2023.

[4] Y. Wang et al. Guided MRI reconstruction via Schrödinger bridge. arXiv:2411.14269, 2024.